

```
1
2 CREATE DATABASE bookflo_db;
3 USE bookflo_db;
4 CREATE TABLE Books
5 (
6     book_id INT AUTO_INCREMENT PRIMARY KEY,
7     title VARCHAR(200) NOT NULL,
8     author VARCHAR(150) NOT NULL,
9     genre VARCHAR(100),
10    description TEXT,
11    published_year INT
12);
13 INSERT INTO Books
14 (title, author, genre, description, published_year)
15 VALUES
16 ('The Martian', 'Andy Weir', 'Science Fiction', 'An astronaut struggles to survive alone on Mars.', 2011),
17 ('Dune', 'Frank Herbert', 'Science Fiction', 'A story involving planets, politics, survival and space.', 1965),
18 ('The Hitchhiker's Guide to the Galaxy', 'Douglas Adams', 'Science Fiction', 'A humorous adventure across space and alien worlds.', 1979),
19 ('Pride and Prejudice', 'Jane Austen', 'Romance', 'A classic story about relationships and social expectations.', 1813),
20 ('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'A story of ambition, wealth, love and the American dream.', 1925),
21 ('Foundation', 'Isaac Asimov', 'Science Fiction', 'A futuristic story about mathematics, civilization and space.', 1951),
22 ('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 'A fantasy adventure involving Bilbo Baggins and a dangerous journey.', 1937),
23 ('Enders Game', 'Orson Scott Card', 'Science Fiction', 'A young student is trained for an interstellar conflict.', 1985),
24 ('Harry Potter and the Sorcerer's Stone', 'J.K. Rowling', 'Fantasy', 'A young wizard begins his magical education.', 1997),
25 ('Twenty Thousand Leagues Under the Sea', 'Jules Verne', 'Adventure', 'A fantastic underwater adventure aboard a mysterious submarine.', 1870);
```

```
19 ('Pride and Prejudice', 'Jane Austen', 'Romance', 'A classic story about relationships and social expectations.', 1813),
20 ('The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 'A story of ambition, wealth, love and the American dream.', 1925),
21 ('Foundation', 'Isaac Asimov', 'Science Fiction', 'A futuristic story about mathematics, civilization and space.', 1951),
22 ('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 'A fantasy adventure involving Bilbo Baggins and a dangerous journey.', 1937),
23 ('Enders Game', 'Orson Scott Card', 'Science Fiction', 'A young student is trained for an interstellar conflict.', 1985),
24 ('Harry Potter and the Sorcerer's Stone', 'J.K. Rowling', 'Fantasy', 'A young wizard begins his magical education.', 1997),
25 ('Twenty Thousand Leagues Under the Sea', 'Jules Verne', 'Adventure', 'A fantastic underwater adventure aboard a mysterious submarine.', 1870);
26 SELECT * FROM Books;
27 ALTER TABLE Books
28 ADD COLUMN embedding JSON;
29 UPDATE Books SET embedding = JSON_ARRAY(0.90,0.80,0.20) WHERE book_id = 1;
30 UPDATE Books SET embedding = JSON_ARRAY(0.85,0.75,0.25) WHERE book_id = 2;
31 UPDATE Books SET embedding = JSON_ARRAY(0.80,0.70,0.30) WHERE book_id = 3;
32 UPDATE Books SET embedding = JSON_ARRAY(0.10,0.20,0.90) WHERE book_id = 4;
33 UPDATE Books SET embedding = JSON_ARRAY(0.15,0.30,0.80) WHERE book_id = 5;
34 UPDATE Books SET embedding = JSON_ARRAY(0.95,0.65,0.20) WHERE book_id = 6;
35 UPDATE Books SET embedding = JSON_ARRAY(0.30,0.75,0.90) WHERE book_id = 7;
36 UPDATE Books SET embedding = JSON_ARRAY(0.80,0.85,0.15) WHERE book_id = 8;
37 UPDATE Books SET embedding = JSON_ARRAY(0.20,0.55,0.95) WHERE book_id = 9;
38 UPDATE Books SET embedding = JSON_ARRAY(0.70,0.90,0.40) WHERE book_id = 10;
39 SELECT book_id, title, embedding;
```

```
37 UPDATE Books SET embedding = JSON_ARRAY(0.20,0.55,0.95) WHERE book_id = 9;
38 UPDATE Books SET embedding = JSON_ARRAY(0.70,0.90,0.40) WHERE book_id = 10;
39 SELECT book_id, title, embedding
40 FROM Books
41 ORDER BY book_id;
42 SELECT
43     book_id,
44     title,
45     author,
46     genre,
47     ROUND(
48         SQRT(
49             POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6))-0.90,2)
50             +
51             POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6))-0.80,2)
52             +
53             POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6))-0.20,2)
54         ),4
55     ) AS l2_distance
56 FROM Books
```

```

61 book_id,
62 title,
63 author,
64 genre,
65 ROUND(
66 (
67 CAST(JSON_EXTRACT(embedding,'$[0]') AS DECIMAL(10,6))*0.90
68 +
69 CAST(JSON_EXTRACT(embedding,'$[1]') AS DECIMAL(10,6))*0.80
70 +
71 CAST(JSON_EXTRACT(embedding,'$[2]') AS DECIMAL(10,6))*0.20
72 )
73 /
74 (
75 Sqrt(
76 Pow(CAST(JSON_EXTRACT(embedding,'$[0]') AS DECIMAL(10,6)),2)
77 +
78 Pow(CAST(JSON_EXTRACT(embedding,'$[1]') AS DECIMAL(10,6)),2)
79 +
80 Pow(CAST(JSON_EXTRACT(embedding,'$[2]') AS DECIMAL(10,6)),2)

```

```

79 +
80 Pow(CAST(JSON_EXTRACT(embedding,'$[2]') AS DECIMAL(10,6)),2)
81 )
82 *
83 Sqrt(Pow(0.90,2)+Pow(0.80,2)+Pow(0.20,2))
84 ),4
85 ) AS cosine_similarity
86 FROM Books
87 WHERE embedding IS NOT NULL
88 ORDER BY cosine_similarity DESC
89 LIMIT 3;
90 * SELECT
91 book_id,
92 title,
93 author,
94 genre,
95 published_year,
96 embedding
97 FROM Books
98 ORDER BY book_id;

```

```

1
2 CREATE DATABASE bookflo_db;
3 USE bookflo_db;
4
5 CREATE TABLE Books
6 (
7 book_id INT AUTO_INCREMENT PRIMARY KEY,
8 title VARCHAR(200) NOT NULL,
9 author VARCHAR(150) NOT NULL,

```

book_id	title	author	genre	description	published_year
1	The Martian	Andy Weir	Science Fiction	An astronaut struggles to survive alone on Mars.	2011
2	Dune	Frank Herbert	Science Fiction	A story involving planets, politics, survival and s...	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	A humorous adventure across space and alien ...	1979
4	Pride and Prejudice	Jane Austen	Romance	A classic story about relationships and social ex...	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	A story of ambition, wealth, love and the Ameri...	1925
6	Foundation	Isaac Asimov	Science Fiction	A futuristic story about mathematics, civilization...	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	A fantasy adventure involving Bilbo Baggins an...	1937
8	Enders Game	Orson Scott Card	Science Fiction	A young student is trained for an interstellar co...	1985
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	A young wizard begins his magical education.	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	A fantastic underwater adventure aboard a my...	1870

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Query 1 SQL File 3* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 11* x

Limit to 1000 rows

```
1
2 • CREATE DATABASE bookflo_db;
3 • USE bookflo_db;
4
5 • CREATE TABLE Books
6 (
7     book_id INT AUTO_INCREMENT PRIMARY KEY,
8     title VARCHAR(200) NOT NULL,
9     author VARCHAR(150) NOT NULL,
```

Result Grid

	book_id	title	embedding
▶	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
	10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

Books 1 Books 2 x Result 3 Result 4 Books 5

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Query 1 SQL File 3* SQL File 5* SQL File 6* SQL File 7* SQL File 8* SQL File 9* SQL File 10* SQL File 11* SQL File 11* x

Limit to 1000 rows

```
1
2 • CREATE DATABASE bookflo_db;
3 • USE bookflo_db;
4
5 • CREATE TABLE Books
6 (
7     book_id INT AUTO_INCREMENT PRIMARY KEY,
8     title VARCHAR(200) NOT NULL,
9     author VARCHAR(150) NOT NULL,
```

Result Grid

	book_id	title	author	genre	published_year	embedding
	2	Dune	Frank Herbert	Science Fiction	1965	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	Jane Austen	Romance	1813	[0.10, 0.20, 0.90]
	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925	[0.15, 0.30, 0.80]
	6	Foundation	Isaac Asimov	Science Fiction	1951	[0.95, 0.65, 0.20]
	7	The Hobbit	J.R.R. Tolkien	Fantasy	1937	[0.30, 0.75, 0.90]
	8	Enders Game	Orson Scott Card	Science Fiction	1985	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997	[0.20, 0.55, 0.95]
	10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870	[0.70, 0.90, 0.40]

Books 1 Books 2 Result 3 Result 4 Books 5 x